Arithmetic and Geometric Sequence Exercises

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§1.1 The trigonometric example and logarithmic progressions

The first exercise concerns a triangle ABC such that

$$\lg(\sin A), \quad \lg(\sin B), \quad \lg(\sin C)$$

form an arithmetic progression, and A, B, C themselves also form an arithmetic progression. Since

$$A + B + C = \pi,$$

if A, B, C are in arithmetic progression, then

$$B = \frac{\pi}{3}.$$

The logarithmic condition means

$$2\lg(\sin B) = \lg(\sin A) + \lg(\sin C),$$

so

$$\sin^2 B = \sin A \sin C.$$

Because $\sin B = \sqrt{3}/2$, we get

$$\sin A \sin C = \frac{3}{4}.$$

Using $A + C = 2\pi/3$, one can analyze

$$\cos(A - C) = \cos A \cos C + \sin A \sin C.$$

The conclusion can be proved without guessing symmetry. Using the product-to-sum identity,

$$2\sin A \sin C = \cos(A - C) - \cos(A + C) = \cos(A - C) + \frac{1}{2}.$$

Since $\sin A \sin C = \frac{3}{4}$, it follows that $\cos(A - C) = 1$. As $-\pi < A - C < \pi$, we have $A = C$, and the triangle is equilateral: $A = B = C = \pi/3$. The decisive step is the exact identity, together with the allowed interval for angle differences.

If

$$a_n = 10 + \lg 2^n,$$

then

$$a_n = 10 + n \lg 2.$$

Hence

$$a_{n+1} - a_n = \lg 2,$$

so $\{a_n\}$ is an arithmetic progression.

This is a typical use of logarithms: they turn powers into multiples. A geometric pattern in 2^n becomes an arithmetic pattern after applying lg.

§1.2 Term recovery by the finite difference of sums

If S_n is the sum of the first n terms of a sequence, then

$$a_n = S_n - S_{n-1}.$$

This applies to problems involving two arithmetic progressions $\{a_n\}$, $\{b_n\}$ whose sums S_n, T_n satisfy

$$\frac{S_n}{T_n} = \frac{2n}{3n+1}.$$

For arithmetic progressions,

$$S_n = \frac{n}{2}(a_1 + a_n),$$

but the cleaner trick is to use

$$a_n = S_n - S_{n-1}, \qquad b_n = T_n - T_{n-1}.$$

One can derive the answer. Since arithmetic progressions have quadratic partial sums, write $S_n = n(A + Bn)$ and $T_n = n(C + Dn)$. Then $(A + Bn)(3n + 1) = 2n(C + Dn)$ for every positive integer n. Comparing coefficients yields $A = 0$, $B = 2C$, $D = 3C$. Taking consecutive differences gives $a_n = 2C(2n - 1)$ and $b_n = 2C(3n - 1)$, so

$$\boxed{\frac{a_n}{b_n} = \frac{2n-1}{3n-1}}.$$

This calculation also requires $T_n \neq 0$, as assumed in the quotient.

A common type of problem gives a relation between S_n and a_n. For example,

$$S_n = 2a_n - 2.$$

Then

$$a_{n+1} = S_{n+1} - S_n = 2a_{n+1} - 2 - (2a_n - 2),$$

so

$$a_{n+1} = 2a_n.$$

Thus the sequence is geometric. The first term is fixed as well: $S_1 = a_1 = 2a_1 - 2$ implies $a_1 = 2$. Therefore the unique sequence is $a_n = 2^n$ for $n \geq 1$. Whenever a problem relates a term to a partial sum, subtract consecutive equations and then check the initial condition.

If $S_n = a_1 + \cdots + a_n$, then

$$a_n = S_n - S_{n-1} = \Delta S_{n-1}.$$

So recovering terms from partial sums is finite differentiation. Conversely,

$$S_n = S_0 + \sum_{k=1}^{n} a_k$$

is finite integration.

For an arithmetic progression $a_n = a_1 + (n-1)d$, the first difference is constant:

$$\Delta a_n = d.$$

For a quadratic sequence, the second difference is constant. This is why finite-difference tables detect polynomial sequences, just as derivatives detect polynomial degree in calculus.

§1.3 Geometric sequences, sums, and telescoping

If $\{a_n\}$ is a geometric progression, then

$$a_m^2 = a_{m-r}a_{m+r}.$$

This appears in several exercises. It follows from

$$a_n = a_1 q^{n-1}.$$

For example,

$$a_6^2 = a_5 a_7.$$

Such identities compute missing terms and ratios from partial information.

Two sums with visually similar denominators behave quite differently. For a product denominator, partial fractions give

$$\sum_{k=1}^{n} \frac{1}{k(k+1)} = \sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 - \frac{1}{n+1}.$$

For an additive denominator, the corresponding sum is

$$\sum_{k=1}^{n} \frac{1}{k+(k+1)} = \sum_{k=1}^{n} \frac{1}{2k+1} = H_{2n+1} - 1 - \frac{1}{2}H_n, \qquad H_m = \sum_{j=1}^{m} \frac{1}{j}.$$

Here the odd-denominator terms are isolated by subtracting the even terms from the harmonic sum. No analogous termwise telescoping occurs. The two formulas illustrate why the denominator's algebraic factorization, rather than its visual appearance, determines which summation technique applies.

A telescoping sum has the form

$$a_k = F(k+1) - F(k).$$

Then

$$\sum_{k=m}^{n} a_k = F(n+1) - F(m).$$

Only the boundary terms survive. In this language, a telescoping term is a discrete derivative or coboundary.

The identity

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}$$

is exactly this pattern with $F(k) = 1/k$. The cancellation is not accidental; it is the discrete fundamental theorem of calculus.

§1.4 Generating functions and second-order recurrences

A sequence (a_n) can be packaged into a generating function

$$A(x) = \sum_{n \geq 0} a_n x^n.$$

Arithmetic and geometric progressions have especially simple forms. For example,

$$\sum_{n \geq 0} q^n x^n = \frac{1}{1 - qx}.$$

A linear sequence produces a rational generating function with denominator $(1-x)^2$. Recurrences translate into algebraic equations for generating functions.

The standard generating-function and characteristic-root treatments may be found in MIT 6.042J, *Mathematics for Computer Science*, Chapters 10–12.[1] For an arithmetic sequence indexed from zero, differentiating $\sum_{n\geq 0} z^n = 1/(1-z)$ shows $\sum_{n\geq 0}(n+1)z^n = (1-z)^{-2}$. Thus the extra factor n corresponds precisely to a repeated denominator factor in the generating function.

Some sequence problems go beyond arithmetic and geometric progressions. A linear recurrence such as

$$a_{n+2} = pa_{n+1} + qa_n$$

is solved by trying $a_n = \lambda^n$. This gives the characteristic equation

$$\lambda^2 = p\lambda + q.$$

When the roots are distinct, the solution has the form

$$a_n = A\lambda_1^n + B\lambda_2^n.$$

Thus geometric progressions are the eigenvectors of the shift recurrence. This is the bridge from elementary sequences to linear algebra and dynamical systems.

[1] MIT 6.042J, *Mathematics for Computer Science*, readings.